ENGINEERING PORTFOLIO

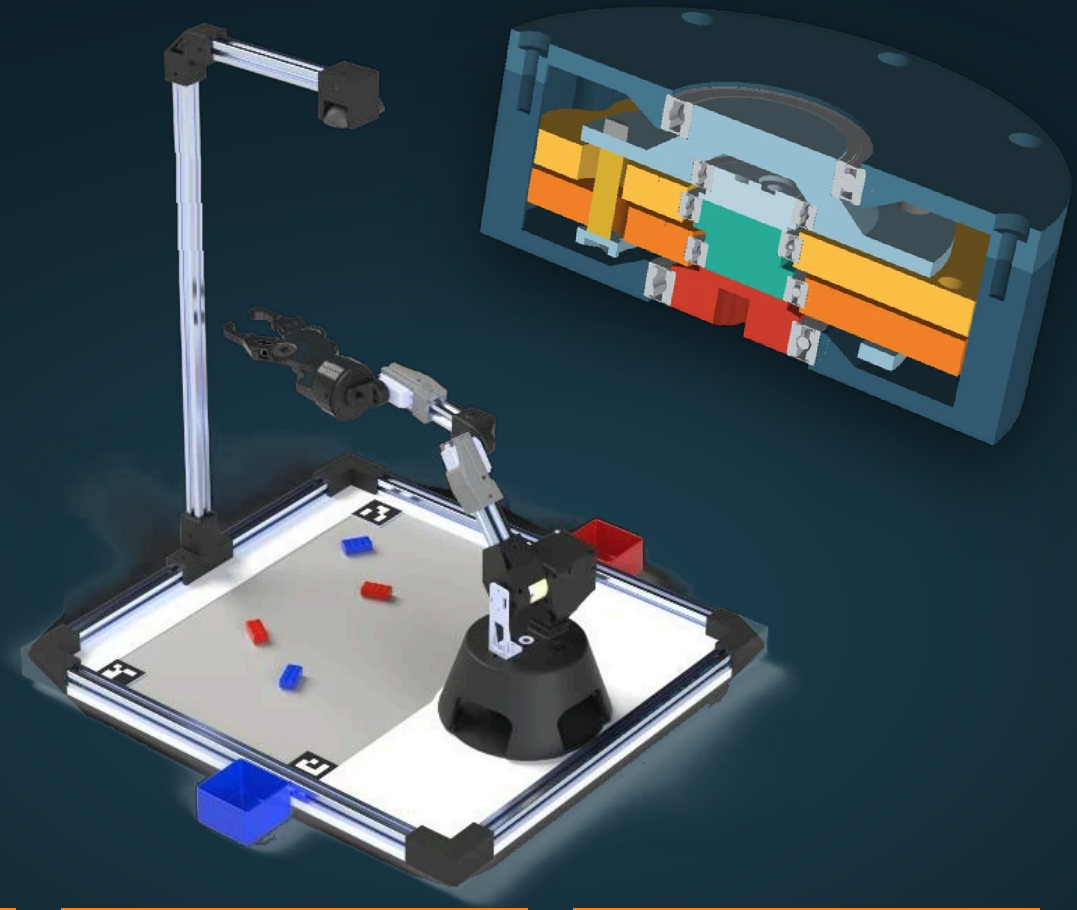
Jamal Charkie

Mechanical engineering student (BEng) at DTU, focused on mechanical design and mechatronics.

I like taking an idea all the way: **calculating it, designing it in SolidWorks and building a prototype** that gets tested and used.

OPEN TO

An engineering internship from February 2027, and a student job alongside my studies.



01 · DTU

ARES-4 modular robot arm

48 / 50

bricks sorted, fully automatic

02 · INDEPENDENT

Cycloidal QDD actuator

15 : 1

printable two-disc reduction

03 · ASSOCIATION

Wall-mounted QR scanner

V2

in use every week

04 · DTU

Motor-driven syringe pump

99–101 %

of the set flow

Also built: a vision-guided nerf turret (YOLOv8n, steppers) · a bike-powered phone charger · a table football goalkeeper bot.

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3D models and video: **jamalch15.github.io/portfolio**

01

COMPLETED DTU · team of 3

ARES-4: a modular robot arm, built for many tasks

Tools swap in seconds, and a browser dashboard sets up and runs new jobs. Tested on **fully automatic LEGO sorting**: find a brick, pick it, drop it in its colour's bin, repeat until the table is clear.

DESIGN	ELECTRONICS	SOFTWARE
SolidWorks · Maple	KiCad · ESP32-S3	Python · FastAPI · vision

48 / 50
bricks sorted with no one touching it

≈500 g
payload carried

≈7 Nm
shoulder peak, for ≈4 Nm needed

WHAT I DID

In a team of three, I drove the **mechanical design, the electronics and the software**: motor sizing and CAD, KiCad wiring, ESP32-S3 firmware, kinematics and the control dashboard.

DESIGN DECISIONS

Strong enough at full reach. About 4 Nm needed at the shoulder; a 27:1 geared stepper gives about 7 Nm peak, confirmed by our torque test.

Light up top. Servos at the elbow and wrist keep the mass near the base.

Quick-swap tools. A 35° twist releases the gripper; a sideways pogo connector powers the tool when it locks.

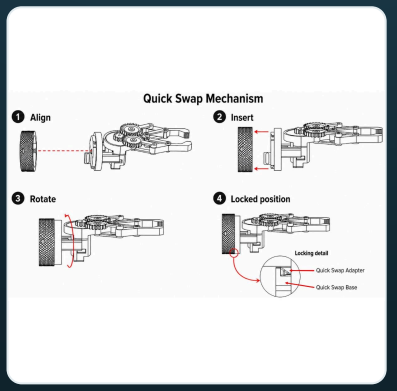
Encoder-corrected shoulder. An AS5048A measures the real angle after each move.



THE ARM IN 3D, FROM THE ORIGINAL SOLIDWORKS ASSEMBLY



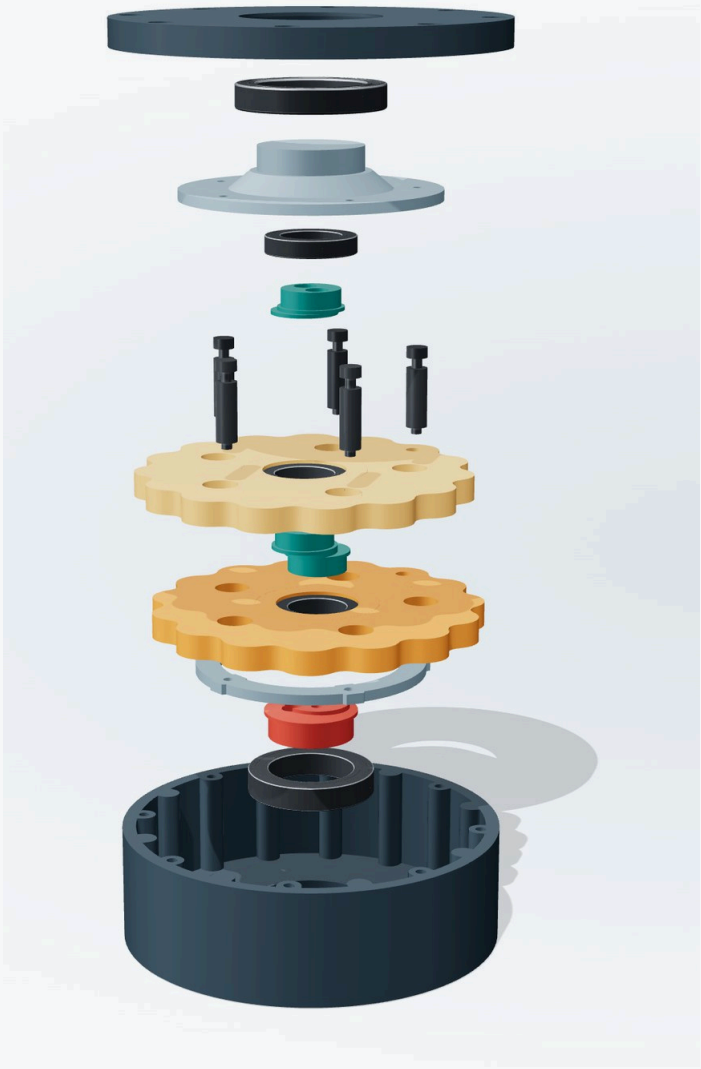
THE REAL CELL



QUICK-SWAP: ALIGN, INSERT, ROTATE, LOCK



DASHBOARD: GHOST ARM AND PLANNED PATH



02

IN DEVELOPMENT Independent project

A printable, backdrivable robot joint

A low-ratio cycloidal drive on a drone motor: the kind of actuator legged robots and cobots use, **designed to be printed at home** around a few bought parts.

DESIGN	DRIVE	BUILD
SolidWorks · cycloids	BLDC · FOC	FDM printing

15 : 1	2 mm	15	180°
reduction, in a 12–16:1 target	eccentricity	lobes per disc	between the discs

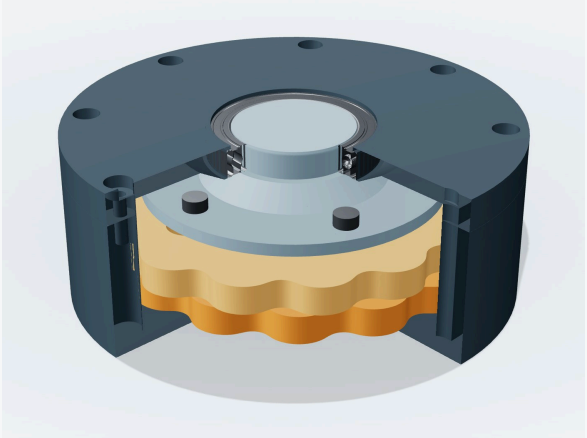
DESIGN DECISIONS

Low ratio on purpose. The joint stays backdrivable and quick to respond.

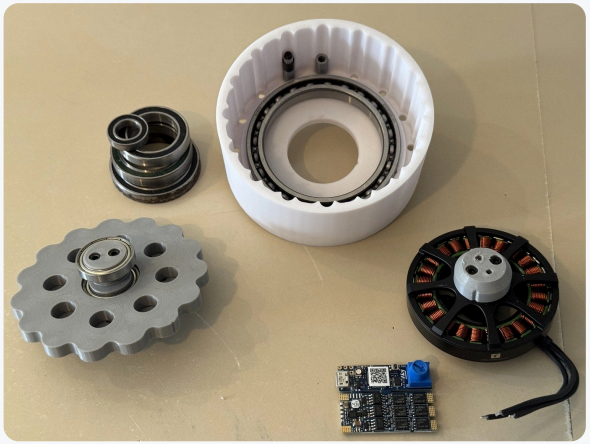
Drone motor, not a stepper. A T-Motor MN6007 II outrunner with field-oriented control.

No ring pins. The discs roll on a profile in the printed housing; printed sleeves over screws had caused seam friction.

Serviceable shaft. A three-piece eccentric shaft clamped with M3 screws, so the bearings fit inside the discs.



CUTAWAY: TWO DISCS INSIDE THE LOBED HOUSING



FIRST PROTOTYPE: PRINTED AND FITTED WITH REAL BEARINGS

Next: fit samples, assembly and hand-turn tests of this two-disc design.

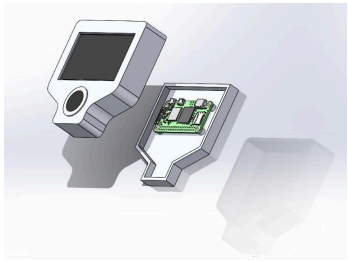
03

IN USE EVERY WEEK · Independent, for an association

QR check-in terminal: built, used, learned from, rebuilt

I designed and built the hardware: the enclosure and wall mount around the display, GM60 scanner and Raspberry Pi. The app software was handled separately.

V1 · ON THE WALL



In use, we **logged its issues** and how people actually used it.

→

V2 · REBUILT TO FIX ALL OF IT

- Larger official 5-inch Raspberry Pi display
- A clearly separate scan opening
- An easier-to-read front
- Simpler wall installation, sturdier mounting
- Quick to assemble and service

In use at the association every week.

DESIGN DECISIONS

Designed around the real parts. The display, scanner and Raspberry Pi set the packaging; the CAD covers fasteners, clearances, cable routing and assembly access.

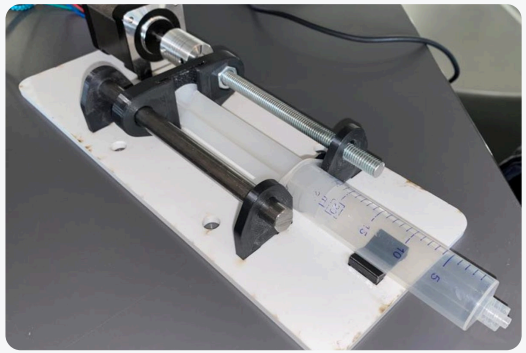
Print-ready. Every housing part is designed for FDM, then printed, assembled and tested on the real hardware.

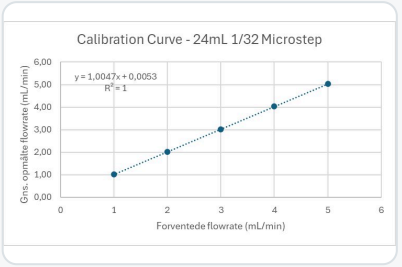
04

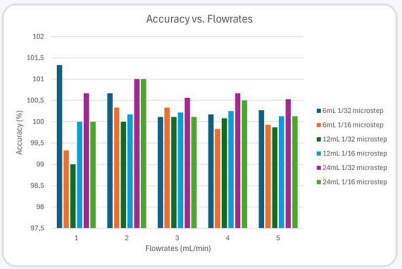
COMPLETED · DTU course project · team of 3

A printed syringe pump within 99–101 % of set flow

A modular pump that pushes and pulls, takes 6–24 mL syringes and turns stepper rotation into precise plunger travel. **I modelled it, built it and measured it.**







99–101 %
of set flow, 6, 12 and 24 mL syringes

1–5
mL/min tested, 3 repeats per point

0.01 g
scale for every measurement

WHAT I DID

CAD modelling (with feedback from the group), **building the prototype**, and the **data collection and processing**.